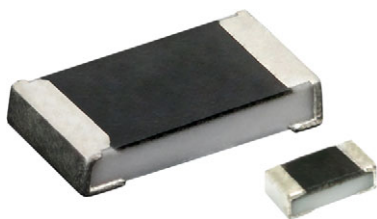


Anti-Surge, High Power Thick Film Chip Resistors



LINKS TO ADDITIONAL RESOURCES



The anti-surge thick film chip resistor series combines a significantly higher power rating and pulse load performance as compared to standard chip resistors.

FEATURES

- Excellent surge pulse capability
- Superior ESD surge characteristics
- High power rating
- AEC-Q200 qualified
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912



RoHS
COMPLIANT
HALOGEN
FREE

APPLICATIONS

- Automotive
- Industrial
- Telecommunications
- Medical

TECHNICAL SPECIFICATIONS				
DESCRIPTION	RCS0402 e3	RCS0603 e3	RCS0805 e3	RCS1206 e3
Imperial size	0402	0603	0805	1206
Metric size code	RR1005M	RR1608M	RR2012M	RR3126M
Resistance range	1 Ω to 10 M Ω ; jumper (0 Ω)			
Resistance tolerance	$\pm 5\%$; $\pm 1\%$; $\pm 0.5\%$			
Temperature coefficient	± 200 ppm/K; ± 100 ppm/K			
Rated dissipation, P_{70} ⁽¹⁾	0.2 W	0.25 W	0.5 W	0.5 W
Rated dissipation, P at terminal part temperature	0.33 W	0.5 W	0.75 W	1 W
	105 $^{\circ}\text{C}$			
Operating voltage, U_{max} AC _{RMS} /DC	50 V	75 V	150 V	200 V
Permissible film temperature, $\vartheta_{\text{F max}}$ ⁽¹⁾	155 $^{\circ}\text{C}$			
Operating temperature range	-55 $^{\circ}\text{C}$ to +155 $^{\circ}\text{C}$			
Max. resistance change at P_{70} for resistance range, $ \Delta R/R $ after:				
	1000 h $\leq 1.0\%$ 8000 h $\leq 2.0\%$			
Permissible voltage against ambient (insulation):				
1 min, U_{ins}	75 V	100 V	200 V	300 V
Failure rate: FIT _{observed}	$\leq 0.1 \times 10^{-9}/\text{h}$			

Note

⁽¹⁾ Please refer to APPLICATION INFORMATION below

APPLICATION INFORMATION

When the resistor dissipates power, a temperature rise above the ambient temperature occurs, dependent on the thermal resistance of the assembled resistor together with the printed circuit board. The rated dissipation applies only if the permitted film temperature is not exceeded.

These resistors do not feature a limited lifetime when operated within the permissible limits. However, resistance value drift increasing over operating time may result in exceeding a limit acceptable to the specific application, thereby establishing a functional lifetime.

**TEMPERATURE COEFFICIENT AND RESISTANCE RANGE**

TYPE / SIZE	TCR	TOLERANCE	RESISTANCE	E-SERIES
RCS0402 e3	± 200 ppm/K	± 5 %	1 Ω to 10 MΩ	E24
	± 100 ppm/K	± 1 %	1 Ω to 10 MΩ	E24; E96
	± 100 ppm/K	± 0.5 %	1 Ω to 10 MΩ	E24; E96
	Jumper, $I_{max.} = 3$ A	≤ 20 mΩ	0 Ω	-
RCS0603 e3	± 200 ppm/K	± 5 %	1 Ω to 10 MΩ	E24
	± 100 ppm/K	± 1 %	1 Ω to 10 MΩ	E24; E96
	± 100 ppm/K	± 0.5 %	1 Ω to 10 MΩ	E24; E96
	Jumper, $I_{max.} = 3.5$ A	≤ 20 mΩ	0 Ω	-
RCS0805 e3	± 200 ppm/K	± 5 %	1 Ω to 10 MΩ	E24
	± 100 ppm/K	± 1 %	1 Ω to 10 MΩ	E24; E96
	± 100 ppm/K	± 0.5 %	1 Ω to 10 MΩ	E24; E96
	Jumper, $I_{max.} = 4$ A	≤ 20 mΩ	0 Ω	-
RCS1206 e3	± 200 ppm/K	± 5 %	1 Ω to 10 MΩ	E24
	± 100 ppm/K	± 1 %	1 Ω to 10 MΩ	E24; E96
	± 100 ppm/K	± 0.5 %	1 Ω to 10 MΩ	E24; E96
	Jumper, $I_{max.} = 5$ A	≤ 20 mΩ	0 Ω	-

Note

- The temperature coefficient of resistance (TCR) is not specified for 0 Ω jumpers

PACKAGING

TYPE / SIZE	CODE	QUANTITY	PACKAGING STYLE	WIDTH	PITCH	PACKAGING DIMENSIONS
RCS0402 e3	ED = ET7 EE = EF4	10 000 50 000	Paper tape according to IEC 60286-3, Type 1a	8 mm	2 mm	Ø 180 mm/7" Ø 330 mm/13"
RCS0603 e3	EI = ET2 ED = ET3 EE = ET8	5000 10 000 50 000			2 mm	Ø 180 mm/7" Ø 180 mm/7" Ø 330 mm/13"
	EA = ET1 EC = ET6	5000 20 000			4 mm	Ø 180 mm/7" Ø 330 mm/13"
	EA = ET1 EC = ET6	5000 20 000			4 mm	Ø 180 mm/7" Ø 330 mm/13"
	EA = ET1 EC = ET6	5000 20 000			4 mm	Ø 180 mm/7" Ø 330 mm/13"
	EA = ET1 EC = ET6	5000 20 000			4 mm	Ø 180 mm/7" Ø 330 mm/13"

PART NUMBER AND PRODUCT DESCRIPTION

Part Number: RCS0805100RFKEC

Part Number: RCS08050000Z0EC

R	C	S	0	8	0	5	1	0	0	R	F	K	E	C
TYPE / SIZE		RESISTANCE		TOLERANCE		TCR		PACKAGING						
RCS0402 RCS0603 RCS0805 RCS1206		R = decimal K = thousand M = million 0000 = jumper		D = ± 0.5 % F = ± 1.0 % J = ± 5.0 % Z = jumper		K = ± 100 ppm/K N = ± 200 ppm/K 0 = jumper		EA, EC, ED, EE, EI						

Product Description: RCS0805 100 100R 1 % ET6 e3

Product Description: RCS0805 0R0 ET6 e3

RCS0805	100	100R	1 %	ET6	e3
TYPE / SIZE	TCR	RESISTANCE	TOLERANCE	PACKAGING	LEAD (Pb)-FREE
RCS0402 RCS0603 RCS0805 RCS1206	± 100 ppm/K ± 200 ppm/K	10R = 10 Ω 100R = 100 Ω 10K = 10 kΩ 1M0 = 1 MΩ 0R0 = jumper	± 0.5 % ± 1.0 % ± 5.0 %	ET1, ET2, ET3, ET6, ET7, ET8, EF4	e3 = pure tin termination finish

DESCRIPTION

Production is strictly controlled and follows an extensive set of instructions established for reproducibility. A cermet film layer and a glass-over are deposited on a high grade (Al_2O_3) ceramic substrate with its prepared inner contacts on both sides. A special laser is used to achieve the target value and the desired power dissipation performance by smoothly fine trimming the resistive layer without damaging the ceramics. The resistor elements are covered by a protective coating designed for electrical, mechanical and climatic protection. The terminations receive a final pure matte tin on nickel plating.

The result of the determined production is verified by an extensive testing procedure on 100 % of the individual chip resistors. Only accepted products are laid directly into the tape in accordance with **IEC 60286-3 Type 1a** ⁽¹⁾.

ASSEMBLY

The resistors are suitable for processing on automatic SMD assembly systems. They are suitable for automatic soldering using wave, reflow or vapor phase as shown in **IEC 61760-1**. The encapsulation is resistant to all cleaning solvents commonly used in the electronics industry, including alcohols, esters and aqueous solutions. The suitability of conformal coatings potting compounds and their processes, if applied, shall be qualified by appropriate means to ensure the long-term stability of the whole system.

The resistors are RoHS-compliant, the pure matte tin plating provides compatibility with lead (Pb)-free and lead-containing soldering processes. Solderability is specified for 2 years after production or requalification. The permitted storage time is 20 years. The immunity of the plating against tin whisker growth has been proven under extensive testing.

MATERIALS

Vishay acknowledges the following systems for the regulation of hazardous substances:

- IEC 62474, Material Declaration for Products of and for the Electrotechnical Industry, with the list of declarable substances given therein ⁽²⁾
- The Global Automotive Declarable Substance List (GADSL) ⁽³⁾
- The REACH regulation (1907/2006/EC) and the related list of substances with very high concern (SVHC) ⁽⁴⁾ for its supply chain

The products do not contain any of the banned substances as per IEC 62474, GADSL, or the SVHC list, see www.vishay.com/how/leadfree.

Hence the products fully comply with the following directives:

- 2000/53/EC End-of-Life Vehicle Directive (ELV) and Annex II (ELV II)
- 2011/65/EU Restriction of the Use of Hazardous Substances Directive (RoHS) with amendment 2015/863/EU
- 2012/19/EU Waste Electrical and Electronic Equipment Directive (WEEE)

Vishay pursues the elimination of conflict minerals from its supply chain, see the Conflict Minerals Policy at www.vishay.com/doc?49037.

APPROVALS

The resistors are qualified according to AEC-Q200.

Where applicable, the resistors are tested in accordance with **EN 140401-802** which refers to **EN 60115-1**, **EN 60115-8** and the variety of environmental test procedures of the **IEC 60068** ⁽¹⁾ series.

RELATED PRODUCTS

For more information about products with superior surge and pulse performance please refer to D/CRCW-IF e3, Pulse Proof Thick Film Chip Resistors datasheet (www.vishay.com/doc?20024).

The CRCW-HP e3 product series is designed for those applications where both enhanced power rating and superior pulse loading performance is required. For ordering CRCW-HP e3 please refer to latest edition of datasheet (www.vishay.com/doc?20043).

For thick film resistors with standard requirements for power rating, please refer to D/CRCW e3, Standard Thick Film Chip Resistors datasheet (www.vishay.com/doc?20035)

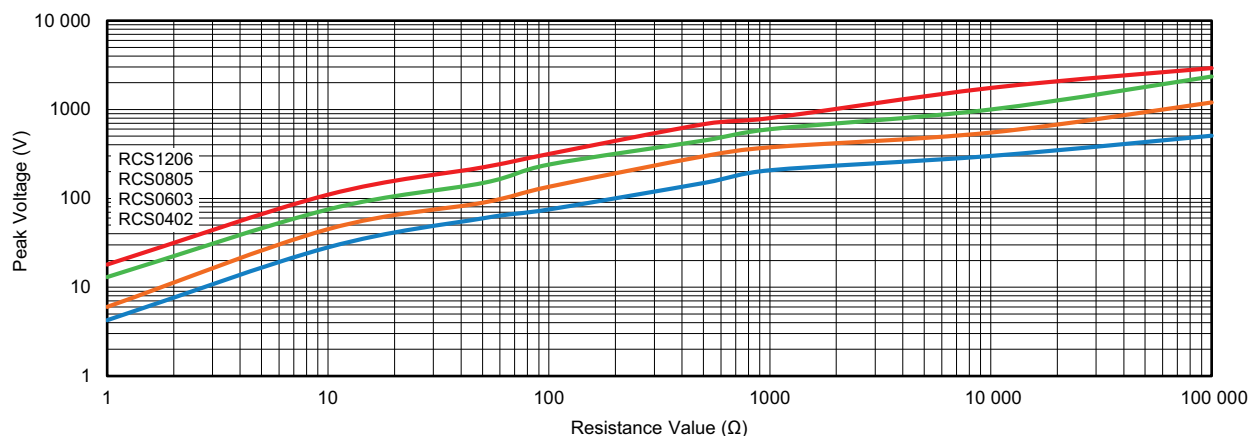
Notes

- (1) The quoted IEC standards are also released as EN standards with the same number and identical contents
- (2) The IEC 62474 list of declarable substances is maintained in a dedicated database, which is available at <http://std.iec.ch/iec62474>.
- (3) The Global Automotive Declarable Substance List (GADSL) is maintained by the American Chemistry Council and available at www.gadsl.org
- (4) The SVHC list is maintained by the European Chemical Agency (ECHA) and available at <http://echa.europa.eu/candidate-list-table>



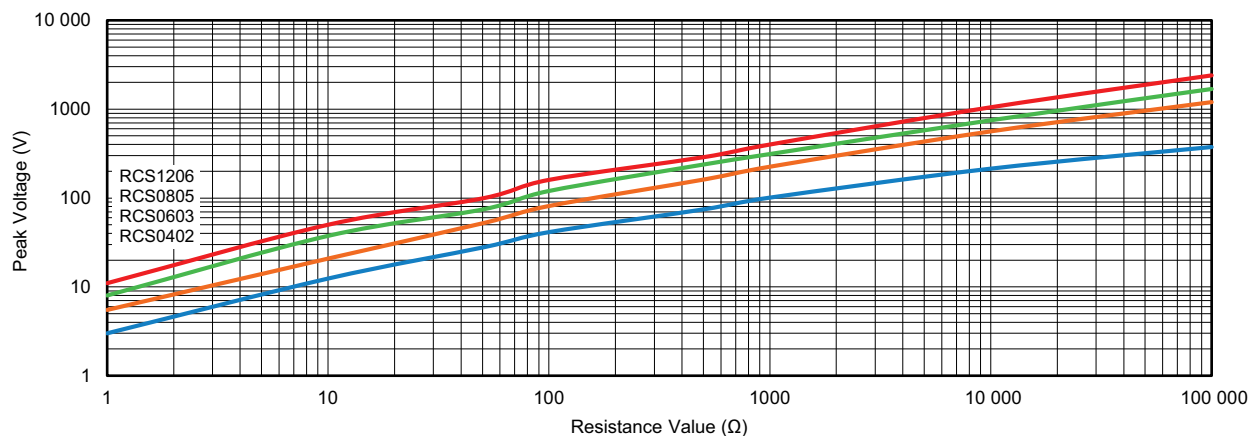
FUNCTIONAL PERFORMANCE

Single Pulse High Power Overload 1.2 μ s / 50 μ s Pulse



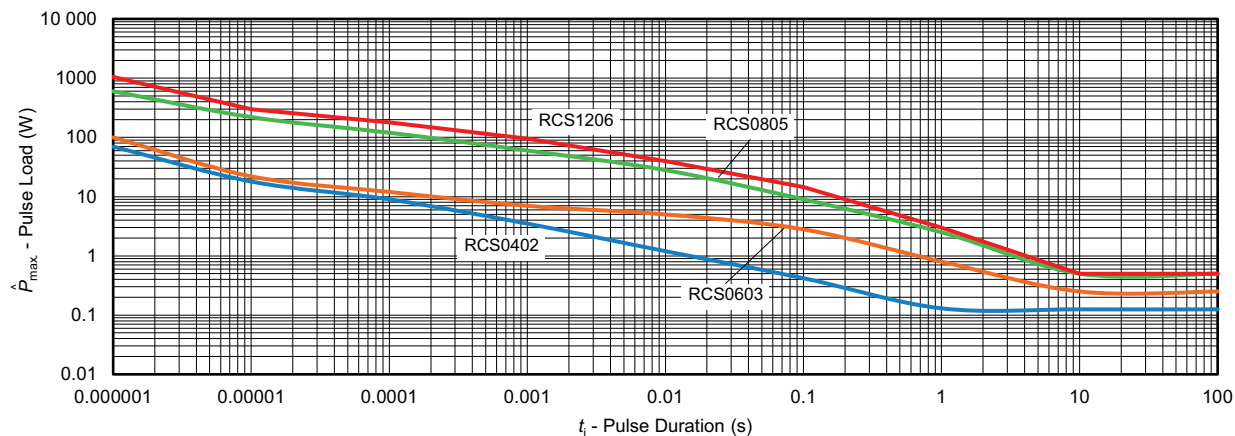
Pulse load rating in accordance with EN 60115-1, 4.27; 1.2 μ s / 50 μ s;
5 pulses at 12 s intervals; for permissible resistance change 1 %

Single Pulse High Power Overload 10 μ s / 700 μ s Pulse

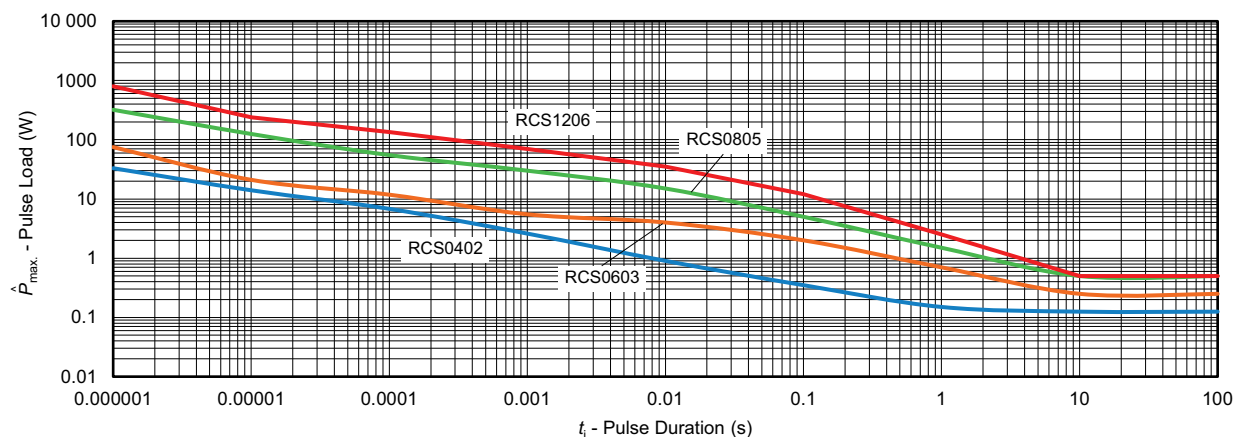


Pulse load rating in accordance with EN 60115-1, 4.27; 10 μ s / 700 μ s;
10 pulses at 1 min intervals; for permissible resistance change 1 %

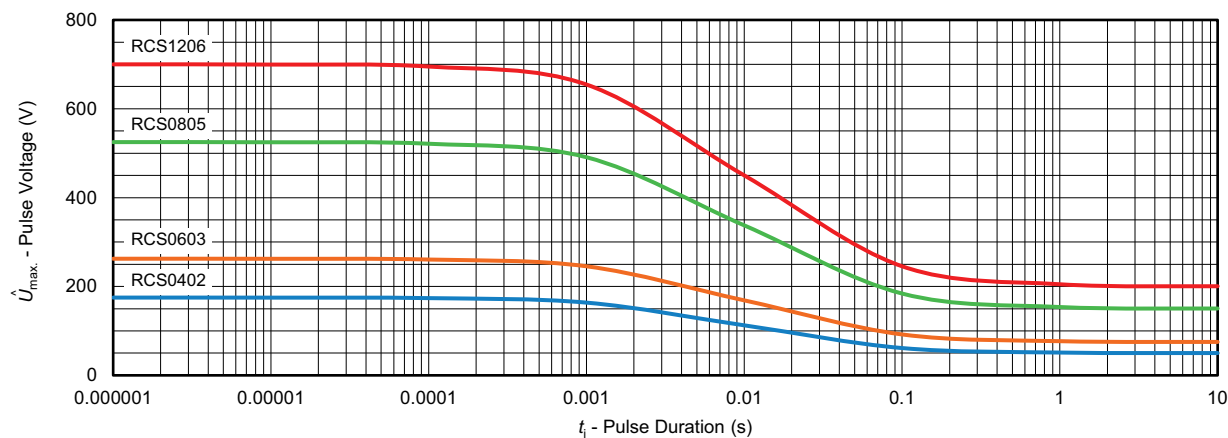
Single Pulse



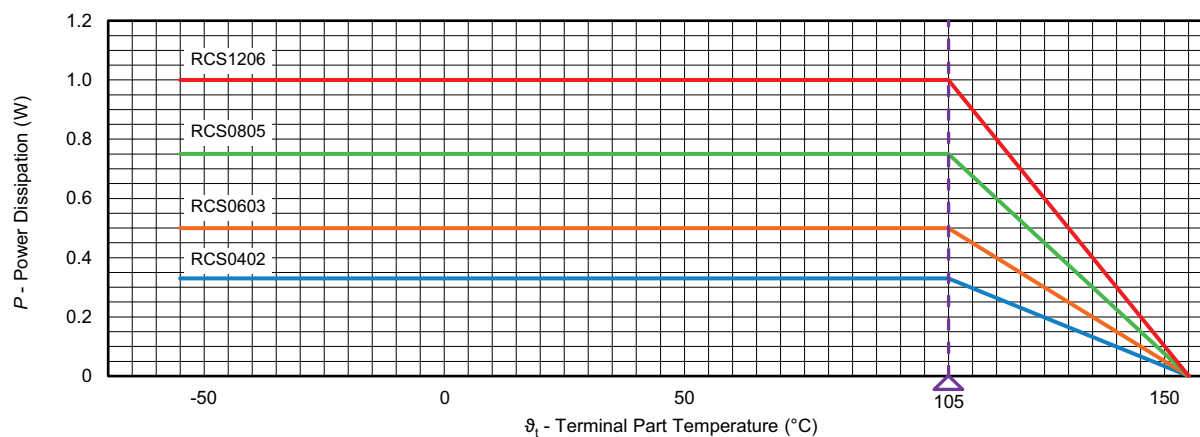
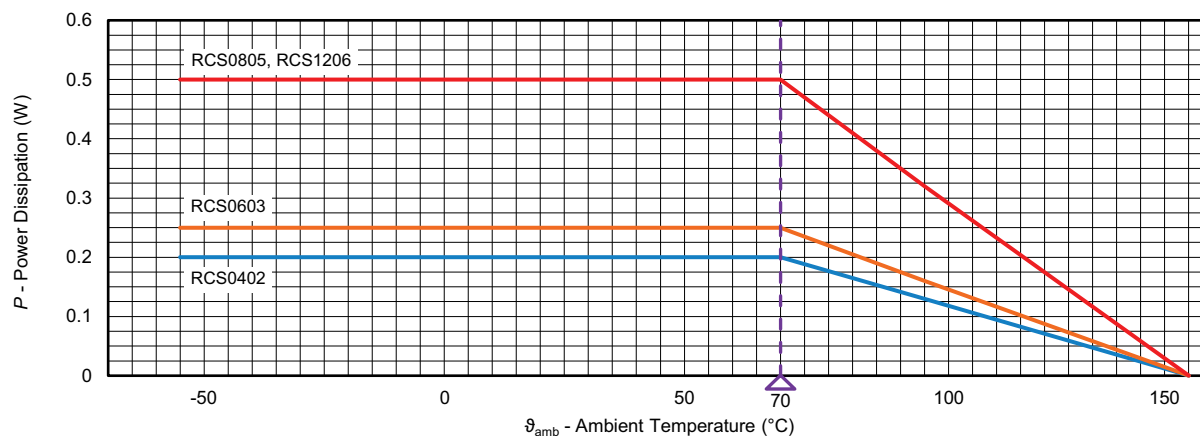
Continuous Pulse



Pulse Voltage



Derating



TESTS AND REQUIREMENTS

All executed tests are carried out in accordance with the following specifications:

EN 60115-1, generic specification

EN 60115-8 (successor of EN 140400), sectional specification

EN 140401-802, detail specification

IEC 60068-2-xx, test methods

The parameters stated in the Test Procedures and Requirements table are based on the required tests and permitted limits of EN 140401-802. The table presents only the most important tests, for the full test schedule refer to the documents listed above. However, some additional tests and a number of improvements against those minimum requirements have been included.

The testing also covers most of the requirements specified by EIA/IS-703 and JIS-C-5201-1.

The tests are carried out under standard atmospheric conditions in accordance with IEC 60068-1, 4.3, whereupon the following values are applied:

Temperature: 15 °C to 35 °C

Relative humidity: 25 % to 75 %

Air pressure: 86 kPa to 106 kPa (860 mbar to 1060 mbar).

A climatic category LCT / UCT / 56 is applied, defined by the lower category temperature (LCT), the upper category temperature (UCT), and the duration of exposure in the damp heat, steady state test (56 days).

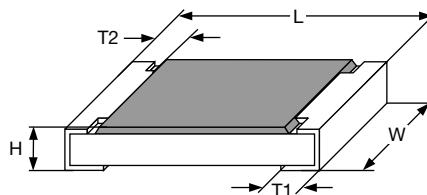
The components are mounted for testing on boards in accordance with EN 60115-8, 2.4.2 unless otherwise specified.

TEST PROCEDURES AND REQUIREMENTS					
EN 60115-1 CLAUSE	IEC 60068-2 (1) TEST METHOD	TEST	PROCEDURE	REQUIREMENTS PERMISSIBLE CHANGE (ΔR)	
			Stability for product types: RCS e3	STABILITY CLASS 1 OR BETTER	STABILITY CLASS 2 OR BETTER
				1 Ω to 10 M Ω	
4.5	-	Resistance	-	$\pm 0.5 \%$; $\pm 1 \%$	$\pm 5 \%$
4.8	-	Temperature coefficient	At (20 / -55 / 20) °C and (20 / 155 / 20) °C	± 100 ppm/K	± 200 ppm/K
4.25.1	-	Endurance at 70 °C	$U = \sqrt{P_{70} \times R}$ or $U = U_{max.}$ whichever is the less severe; 1.5 h on; 0.5 h off 70 °C; 1000 h 70 °C; 8000 h	$\pm (1 \% R + 0.05 \Omega)$ $\pm (2 \% R + 0.1 \Omega)$	$\pm (2 \% R + 0.1 \Omega)$ $\pm (4 \% R + 0.1 \Omega)$
4.25.3	-	Endurance at upper category temperature	155 °C; 1000 h	$\pm (1 \% R + 0.05 \Omega)$	$\pm (2 \% R + 0.1 \Omega)$
4.24	78 (Cab)	Damp heat, steady state	(40 $\pm$ 2) °C; 56 days; (93 $\pm$ 3) % RH	$\pm (1 \% R + 0.05 \Omega)$	
4.37	67 (Cy)	Damp heat, steady state, accelerated	(85 $\pm$ 2) °C; (85 $\pm$ 5) % RH $U = \sqrt{0.1 \times P_{85} \times R} \leq 100$ V; 1000 h	$\pm (1 \% R + 0.05 \Omega)$	$\pm (2 \% R + 0.1 \Omega)$
4.23 4.23.2 4.23.3 4.23.4 4.23.5 4.23.6 4.23.7	- 2 (Bb) 30 (Db) 1 (Ab) 13 (M) 30 (Db) -	Climatic sequence: Dry heat Damp Cold Low air pressure Damp heat, cyclic DC load	125 °C; 16 h 55 °C; 24 h; ≥ 90 % RH; 1 cycle -55 °C; 2 h 8.5 kPa; 2 h; (25 $\pm$ 10) °C 55 °C; 24 h; ≥ 90 % RH; 5 cycles $U = \sqrt{P_{70} \times R} \leq U_{max.}$; 1 min	$\pm (1 \% R + 0.05 \Omega)$	$\pm (2 \% R + 0.1 \Omega)$
-	1 (Aa)	Cold	-55 °C; 2 h	$\pm (0.25 \% R + 0.05 \Omega)$	$\pm (0.5 \% R + 0.05 \Omega)$
4.19	14 (Na)	Rapid change of temperature	30 min. at -55 °C and 30 min. at 125 °C 1000 cycles	$\pm (1 \% R + 0.05 \Omega)$ no visible damage	
4.13	-	Short time overload	$U = 2.5 \times \sqrt{P_{70} \times R} \leq 2 \times U_{max.}$; whichever is the less severe; 5 s	$\pm (2 \% R + 0.05 \Omega)$	

TEST PROCEDURES AND REQUIREMENTS					
EN 60115-1 CLAUSE	IEC 60068-2 ⁽¹⁾ TEST METHOD	TEST	PROCEDURE	REQUIREMENTS PERMISSIBLE CHANGE (ΔR)	
				STABILITY CLASS 1 OR BETTER	STABILITY CLASS 2 OR BETTER
			Stability for product types: RCS e3	1 Ω to 10 M Ω	
4.27	-	Single pulse high voltage overload	Severity no. 4: $U = 10 \times \sqrt{P_{70} \times R}$ or $U = 2 \times U_{max.}$; whichever is the less severe; 10 pulses 10 μ s / 700 μ s	$\pm (1 \% R + 0.05 \Omega)$ no visible damage	
4.39	-	Periodic electric overload	$U = \sqrt{15 \times P_{70} \times R}$ or $U = 2 \times U_{max.}$; whichever is the less severe; 0.1 s on; 2.5 s off; 1000 cycles	$\pm (1 \% R + 0.05 \Omega)$ no visible damage	
4.38	-	Electrostatic discharge (human body model)	IEC 61340-3-1 ⁽¹⁾ ; 3 positive + 3 negative discharges; ESD voltage according to the size	$\pm (1 \% R + 0.05 \Omega)$	
4.22	6 (Fc)	Vibration	Endurance by sweeping; 10 Hz to 2000 Hz; no resonance; amplitude ≤ 1.5 mm or ≤ 200 m/s ² ; 7.5 h	$\pm (0.25 \% R + 0.05 \Omega)$ no visible damage	$\pm (0.5 \% R + 0.05 \Omega)$ no visible damage
4.17	58 (Td)	Solderability	Solder bath method, SnPb40; non-activated flux (235 $\pm$ 5) $^{\circ}$ C; (2 $\pm$ 0.2) s Solder bath method, Sn96.5Ag3Cu0.5; non-activated flux (245 $\pm$ 5) $^{\circ}$ C; (3 $\pm$ 0.3) s	Good tinning (≥ 95 % covered); no visible damage	
4.18	58 (Td)	Resistance to soldering heat	Soldering bath method; (260 $\pm$ 5) $^{\circ}$ C; (10 $\pm$ 1) s	$\pm (0.25 \% R + 0.05 \Omega)$	$\pm (0.5 \% R + 0.05 \Omega)$
4.29	45 (XA)	Component solvent resistance	Isopropyl alcohol +50 $^{\circ}$ C; method 2	No visible damage	
4.32	21 (Ue ₃)	Shear (adhesion)	RCS0402 e3: 9 N RCS0603 e3 to RCS1206 e3: 17.7 N	No visible damage	
4.33	21 (Ue ₁)	Substrate bending	Depth 2 mm; 3 times	$\pm (0.25 \% R + 0.05 \Omega)$ no visible damage, no open circuit in bent position	
4.7	-	Voltage proof	$U = 1.4 \times U_{ins}$; 60 s	No flashover or breakdown	
4.35	-	Flammability, needle flame test	IEC 60695-11-5 ⁽¹⁾ ; 10 s	No burning after 30 s	

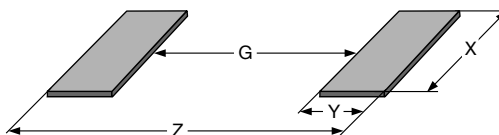
Note
⁽¹⁾ The quoted IEC standards are also released as EN standards with the same number and identical contents

DIMENSIONS



DIMENSIONS AND MASS						
TYPE / SIZE	L (mm)	W (mm)	H (mm)	T1 (mm)	T2 (mm)	MASS (mg)
RCS0402 e3	1.0 ± 0.05	0.5 ± 0.05	0.35 ± 0.05	0.25 ± 0.10	0.2 ± 0.10	0.65
RCS0603 e3	$1.55 + 0.10 / - 0.05$	0.85 ± 0.10	0.45 ± 0.05	0.3 ± 0.20	0.3 ± 0.20	2
RCS0805 e3	$2.0 + 0.20 / - 0.10$	1.25 ± 0.15	0.5 ± 0.10	$0.3 + 0.20 / - 0.10$	0.3 ± 0.20	5.5
RCS1206 e3	$3.2 + 0.10 / - 0.20$	1.6 ± 0.15	0.55 ± 0.10	0.45 ± 0.20	0.4 ± 0.20	10

SOLDER PAD DIMENSIONS



RECOMMENDED SOLDER PAD DIMENSIONS								
TYPE / SIZE	WAVE SOLDERING				REFLOW SOLDERING			
	G (mm)	Y (mm)	X (mm)	Z (mm)	G (mm)	Y (mm)	X (mm)	Z (mm)
RCS0402 e3	-	-	-	-	0.45	0.6	0.6	1.65
RCS0603 e3	0.65	1.10	1.25	2.85	0.75	0.75	1.00	2.25
RCS0805 e3	0.90	1.30	1.60	3.50	1.00	0.95	1.45	2.90
RCS1206 e3	1.40	1.40	1.95	4.20	1.50	1.05	1.80	3.60

Notes

- The given solder pad dimensions reflect the considerations for board design and assembly as outlined e.g in standards IEC 61188-5-x ⁽¹⁾ or in publication IPC-7351
- ⁽¹⁾ The quoted IEC standards are also released as EN standards with the same number and identical contents



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